

1. Inequalities

Markov's Inequality:

$$P(u(X) > c) \leq \frac{E[u(X)]}{c}$$

Chebyshev's Inequality:

$$P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2} \quad \text{or} \quad P(|X - \mu| > k\sigma) \leq \frac{1}{k^2}$$

Proof: $Var(X) = E[(X - \mu)^2]$. Let $u(X) = (X - \mu)^2$. By Markov's: $P((X - \mu)^2 > c) \leq \frac{E[(X - \mu)^2]}{c}$. Let $c = k^2\sigma^2$, then $P(|X - \mu| > k\sigma) \leq \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$.

Jensen's Inequality: If $f''(x) \geq 0$ (convex), then $f(E[X]) \leq E[f(X)]$. If $f''(x) > 0$ (strictly convex), then $f(E[X]) < E[f(X)]$.

2. Transformation & MGF

Moment Generating Function (MGF): $M^{(m)}(0) = E[X^m]$, if $m \in \mathbb{N}$. **Theorem:** $F_X(z) = F_Y(z)$ iff $M_X(t) = M_Y(t)$, $\forall -h < t < h$.

Expectation Existence: If $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$, then $E[X] = \int_{-\infty}^{\infty} xf(x)dx$.

Transformation $Y = g(X)$:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

CDF method ($F_Y(y) = P(g(X) \leq y)$):

- Case 1 ($g \uparrow$): $F_Y(y) = F_X(g^{-1}(y))$
- Case 2 ($g \downarrow$): $F_Y(y) = 1 - F_X(g^{-1}(y))$

Multivariate Transformation: Let $\mathbf{Y} = T(\mathbf{X})$, $\mathbf{X} = T^{-1}(\mathbf{Y})$.

$$f_Y(\mathbf{y}) = f_X(T^{-1}(\mathbf{y})) \cdot |J|, \quad \text{where } J = \det\left(\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right).$$

3. Double Expectation & Bivariate Normal

Double Expectation (Adam's Law):

$$E_{X_1}[E[X_2|X_1]] = E[X_2]$$

Conditional Variance (Eve's Law):

$$Var(X_2) = E[Var(X_2|X_1)] + Var(E[X_2|X_1])$$

Covariance properties: $Cov(X, Y) = E[XY] - \mu_X\mu_Y$. $Var(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2abCov(X, Y)$.

Bivariate Normal (X, Y) : Notation: $(X, Y) \sim N_2(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$.

PDF: $f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{Q}{2(1-\rho^2)}\right\}$

$$Q = \left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2$$

MGF: $\exp(t_1\mu_X + t_2\mu_Y + \frac{1}{2}(t_1^2\sigma_X^2 + t_2^2\sigma_Y^2 + 2\rho t_1t_2\sigma_X\sigma_Y))$

Indep: For Bivariate Normal, Uncorrelated ($\rho = 0$) $\iff$ Independent.

Conditional Normal Distribution: If (X, Y) is Bivariate Normal, then $Y|X = x$ is Normal.

$$E[Y|X = x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X) \quad (\text{Linear Regression})$$

$$Var(Y|X = x) = \sigma_Y^2(1 - \rho^2) \quad (\text{Constant Variance})$$

4. Discrete Distributions

Bernoulli(p): $E = p, Var = p(1 - p)$.

Binomial(n, p): $P(X = k) = \binom{n}{k}p^kq^{n-k}$
 $E = np, Var = npq, M(t) = (pe^t + q)^n$

Geometric(p): (x failures before 1st success) $P(X = x) = p(1 - p)^x, x \in \{0, 1, \dots\}$

$$E[X] = \frac{1-p}{p}, Var(X) = \frac{1-p}{p^2}$$

(Note: If counting trials $n \geq 1$, $E = 1/p, Var = (1 - p)/p^2$)

Negative Binomial(r, p): (y fails before r success) $P(Y = y) = \binom{y+r-1}{y}p^r(1 - p)^y$

$$M_Y(t) = \left(\frac{p}{1-(1-p)e^t}\right)^r, E = \frac{r(1-p)}{p}, Var = \frac{r(1-p)}{p^2}$$

Multinomial: $P(X_1 \dots X_k) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$

$$E[X_i] = np_i, Var(X_i) = np_i(1 - p_i)$$

$$Cov(X_i, X_j) = -np_i p_j$$

$$M(\mathbf{t}) = (\sum p_i e^{t_i})^n$$

Hypergeometric: $P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$

$$E[X] = \frac{nK}{N}, Var(X) = \frac{nK}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$$

Poisson(λ): $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

$$E = \lambda, Var = \lambda, M(t) = \exp(\lambda(e^t - 1))$$

5. Continuous Distributions

Exponential(λ): $f(x) = \lambda e^{-\lambda x}, x \geq 0$

$$E[X] = 1/\lambda, Var(X) = 1/\lambda^2$$

$$F(x) = 1 - e^{-\lambda x}$$

Gamma(α, β): $f(x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}$

$$E = \alpha\beta, Var = \alpha\beta^2, M(t) = (1 - \beta t)^{-\alpha}$$

$\Gamma(\alpha) = (\alpha - 1)!$ if integer.

Chi-Squared(k): ($Gamma(k/2, 2)$) $f(x) = \frac{x^{k/2-1} e^{-x/2}}{2^{k/2} \Gamma(k/2)}$

$$E = k, Var = 2k, M(t) = (1 - 2t)^{-k/2}$$

Beta(α, β): $f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$

$$E[X] = \frac{\alpha}{\alpha+\beta}$$

$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Normal(μ, σ^2): $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$E = \mu, Var = \sigma^2, M(t) = e^{\mu t + \sigma^2 t^2/2}$$

Student's t (v): $T = \frac{Z}{\sqrt{V/v}}$ where $Z \sim N(0, 1), V \sim \chi_v^2$

indep.

$$E[T] = 0 \quad (\text{if } v > 1), Var(T) = \frac{v}{v-2} \quad (\text{if } v > 2).$$

PDF is bell-shaped but heavier tails than Normal.

F Distribution (n, m): $F = \frac{U/n}{V/m}$ where $U \sim \chi_n^2, V \sim \chi_m^2$

indep.

$$E[F] = \frac{m}{m-2} \quad (\text{if } m > 2).$$

Relationship: $T_v^2 \sim F_{1,v}$.

Additive Properties (Indep X_i):

$$X_i \sim Gamma(\alpha_i, \beta) \implies \sum X_i \sim Gamma(\sum \alpha_i, \beta)$$

$$X_i \sim \chi_{k_i}^2 \implies \sum X_i \sim \chi_{\sum k_i}^2$$

$$X_i \sim N(\mu_i, \sigma_i^2) \implies \sum (a_i X_i) \sim N(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2)$$

$$X_i \sim Poisson(\lambda_i) \implies \sum X_i \sim Poisson(\sum \lambda_i)$$

6. Limit Theorems & Estimation

Central Limit Theorem (CLT): $\frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$.

Delta Method (1st & 2nd Order): Let $Y \sim (\theta, \sigma_Y^2)$ and $Y \xrightarrow{P} \theta$.

1. First Order (Variance Approx):

$$g(Y) \approx g(\theta) + g'(\theta)(Y - \theta)$$

$$\implies Var(g(Y)) \approx [g'(\theta)]^2 \sigma_Y^2.$$

2. Second Order (Bias Correction):

$$g(Y) \approx g(\theta) + g'(\theta)(Y - \theta) + \frac{1}{2}g''(\theta)(Y - \theta)^2$$

$$\text{Taking expectation } (E[Y - \theta] = 0, E[(Y - \theta)^2] = \sigma_Y^2):$$

$$\implies E[g(Y)] \approx g(\theta) + \frac{1}{2}g''(\theta)\sigma_Y^2.$$

(Note: If $g''(\theta) \neq 0$, $g(\bar{X})$ is a biased estimator of $g(\theta)$).

Weak Law of Large Numbers (WLLN): $\bar{X}_n \xrightarrow{P} \mu$. Proof

by Chebyshev: $P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0$.

Strong Law of Large Numbers (SLLN): Kolmogorov: $\bar{X}_n \xrightarrow{a.s.} \mu$ iff $E[|X_i|] < \infty$. $P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$.

Maximum Likelihood (MLE): $L(\theta) = \prod f(x_i; \theta)$. Maximize $\ln L(\theta)$.

Property: $\hat{\theta}_{MLE} \xrightarrow{P} \theta_0$, $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow N(0, I(\theta_0)^{-1})$.

Order Statistics: $g(y_1 \dots y_n) = n! \prod f(y_i)$, $a < y_1 < \dots < y_n < b$

$$g_k(y_k) = \frac{n!}{(k-1)!(n-k)!} [F(y_k)]^{k-1} [1 - F(y_k)]^{n-k} f(y_k)$$

$$f_{\min}(y) = n[1 - F(y)]^{n-1} f(y)$$

$$f_{\max}(y) = n[F(y)]^{n-1} f(y)$$

7. Confidence Intervals (Chapter 4)

Mean (μ):

- σ^2 known: $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
- σ^2 unknown: $\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$

Difference of Means ($\mu_1 - \mu_2$):

- $\sigma_{1,2}^2$ known: $(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
- $\sigma_{1,2}^2$ unknown (n large): use S_1, S_2 replace σ .
- Equal Var $\sigma_1^2 = \sigma_2^2$: $(\bar{X} - \bar{Y}) \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
where $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$

Proportion (p): (Large Sample) $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Variance (σ^2): (Normal Population) $\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right)$

8. Convergence & Asymptotics

Relations: $X_n \xrightarrow{P} X \implies X_n \xrightarrow{D} X$.

$X_n \xrightarrow{D} c \iff X_n \xrightarrow{P} c$ (constant c).

Slutsky's Theorem: If $X_n \xrightarrow{D} X$ and $Y_n \xrightarrow{P} c$, then:

$$X_n + Y_n \xrightarrow{D} X + c, \quad X_n Y_n \xrightarrow{D} cX.$$

Continuous Mapping Theorem: If $X_n \xrightarrow{P} X$ and g is continuous, then $g(X_n) \xrightarrow{P} g(X)$. (Also holds for $\xrightarrow{D}$).

Fisher Information: $I(\theta) = E \left[\left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right)^2 \right] = -E \left[\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right]$

Asymptotic Normality of MLE:

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) \xrightarrow{D} N(0, I_1(\theta_0)^{-1})$$

Estimation of Fisher Information: Since θ_0 is unknown, we estimate $I(\theta_0)$ using $\hat{\theta}_{MLE}$:

1. Expected Fisher Information (at $\hat{\theta}$):

$$\hat{I}_n(\hat{\theta}) = nI_1(\hat{\theta}) = nE \left[\left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right)^2 \right]_{\theta=\hat{\theta}}$$

2. Observed Fisher Information:

$$J_n(\hat{\theta}) = - \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \ln f(X_i; \theta) \Big|_{\theta=\hat{\theta}}$$

Result: For large n , both provide approximations for variance:

$$Var(\hat{\theta}) \approx \frac{1}{\hat{I}_n(\hat{\theta})} \approx \frac{1}{J_n(\hat{\theta})}$$

(Observed Info is often preferred in practice, Efron & Hinkley).

Chi-Square Test for Independence (Q_{ab-1}): For contingency table with a rows and b cols.

$H_0 : P(A_i \cap B_j) = P(A_i)P(B_j)$ (Independence).

$$Q_{ab-1} = \sum_{j=1}^b \sum_{i=1}^a \frac{(X_{ij} - n\hat{p}_{i.}\hat{p}_{.j})^2}{n\hat{p}_{i.}\hat{p}_{.j}} \xrightarrow{D} \chi_{(a-1)(b-1)}^2$$

Estimates:

$$\hat{p}_{i.} = \frac{X_{i.}}{n} = \frac{\sum_{j=1}^b X_{ij}}{n} \text{ (Row marginal prob)}$$

$$\hat{p}_{.j} = \frac{X_{.j}}{n} = \frac{\sum_{i=1}^a X_{ij}}{n} \text{ (Col marginal prob)}$$

Note: $n\hat{p}_{i.}\hat{p}_{.j}$ is the Expected Frequency (E_{ij}).

9. Proof of Slutsky's Theorem

Statement: If $X_n \xrightarrow{D} X$ and $Y_n \xrightarrow{P} c$, then $X_n + Y_n \xrightarrow{D} X + c$.

Proof: Let F be the CDF of X . We show $F_{X_n+Y_n}(z) \rightarrow F_X(z-c)$. For any $\epsilon > 0$:

1. Upper Bound:

Decompose $\{X_n + Y_n \leq z\}$:

$$P(X_n + Y_n \leq z) \leq P(X_n + Y_n \leq z, |Y_n - c| < \epsilon) + P(|Y_n - c| \geq \epsilon)$$

Note that if $|Y_n - c| < \epsilon$, then $Y_n > c - \epsilon$.

$$\implies X_n \leq z - Y_n \leq z - c + \epsilon.$$

$$\text{Thus, } P(X_n + Y_n \leq z) \leq P(X_n \leq z - c + \epsilon) + P(|Y_n - c| \geq \epsilon).$$

Taking $\limsup_{n \rightarrow \infty}$ (using $Y_n \xrightarrow{P} c$):

$$\limsup P(X_n + Y_n \leq z) \leq F_X(z - c + \epsilon).$$

2. Lower Bound:

Consider event $\{X_n \leq z - c - \epsilon\}$:

$$P(X_n \leq z - c - \epsilon) \leq P(X_n + Y_n \leq z) + P(|Y_n - c| \geq \epsilon).$$

Taking $\liminf_{n \rightarrow \infty}$:

$$F_X(z - c - \epsilon) \leq \liminf P(X_n + Y_n \leq z).$$

3. Conclusion:

Let $\epsilon \downarrow 0$. Since $z - c$ is a continuity point of F_X :

$$F_X(z - c) \leq \liminf \leq \limsup \leq F_X(z - c).$$

$$\therefore X_n + Y_n \xrightarrow{D} X + c. \quad \square$$

10. Boundedness & Little-o (Thm 5.2.7, 5.2.8)

Def: Sequence X_n is *bounded in probability* ($X_n = O_p(1)$) if $\forall \epsilon > 0, \exists B_\epsilon, N_\epsilon$ such that $P(|X_n| \leq B_\epsilon) \geq 1 - \epsilon$ for all $n \geq N_\epsilon$.

(Note: $X_n \xrightarrow{D} X \implies X_n$ is bounded in prob.)

Theorem 5.2.7 (Product with $\xrightarrow{P} 0$): If X_n is bounded in probability and $Y_n \xrightarrow{P} 0$, then $X_n Y_n \xrightarrow{P} 0$.

Proof: For any $\epsilon > 0$, choose B s.t. $P(|X_n| > B) < \epsilon/2$.

$$P(|X_n Y_n| \geq \epsilon) \leq P(|X_n Y_n| \geq \epsilon, |X_n| \leq B) + P(|X_n| > B) \\ \leq P(B|Y_n| \geq \epsilon) + \epsilon/2 = P(|Y_n| \geq \epsilon/B) + \epsilon/2.$$

Since $Y_n \xrightarrow{P} 0$, the first term $\rightarrow 0$. Thus $\limsup P(\dots) \leq \epsilon/2$. $\square$

Theorem 5.2.8 (o_p Convergence): If Y_n is bounded in probability and $X_n = o_p(Y_n)$ (i.e., $X_n/Y_n \xrightarrow{P} 0$), then $X_n \xrightarrow{P} 0$.

Proof: Let $\epsilon > 0$. Choose B s.t. $P(|Y_n| \leq B) \geq 1 - \epsilon/2$.

$$P(|X_n| \geq \epsilon) = P\left(\left|\frac{X_n}{Y_n} \cdot Y_n\right| \geq \epsilon\right) \\ \leq P\left(\left|\frac{X_n}{Y_n}\right| \cdot B \geq \epsilon\right) + P(|Y_n| > B) \\ = P\left(\left|\frac{X_n}{Y_n}\right| \geq \frac{\epsilon}{B}\right) + \epsilon/2.$$

Since $\frac{X_n}{Y_n} \xrightarrow{P} 0$, the first term $\rightarrow 0$. $\square$